THE CYBER ASSETS IN VITAL SECTORS FACED CONCERTED ATTACKS BY HACKERS FROM PAKISTAN AND OTHER NATIONS BEFORE, DURING AND AFTER OPERATION SINDOOR. ALMOST ALL WERE REPULSED, BUT INDIA CANNOT LET ITS GUARD DOWN

By PRADIP R. SAGAR

Twenty-first century warfare isn't limited to ground, air and naval forces. With vital domains like defence, finance and communications dependent on sets of interconnected information systems on the internet, attacking these can grievously impair a nation's war-making capabilities. Cyber warfare has thus emerged as a low-cost weapon of modern conflict and cyber security is now an important factor in the national security matrix of every state. As in Russia's war on Ukraine and the recent Israel-Iran war, this whole spectrum played out during the recent face-off between India and Pakistan.

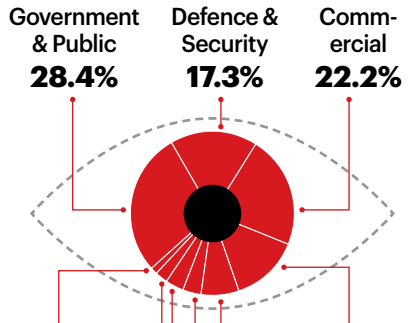
Starting from the Pahalgam terror attack on April 22 and continuing through Operation Sindoor and the four-day conflict between the two neighbours (May 7-10), India faced an unrelenting wave of cyberattacks, primarily by Pakistani groups backed by Islamabad and Beijing, but also from hacker groups in Turkey, Bangladesh, Malaysia and West Asian countries. According to Indian government sources, these groups launched over 1.5 million cyber attacks targeting a wide swathe of India's critical infrastructure spanning defence, power, telecom, finance and transportation during this period. In a cabinet meeting in early June, Prime Minister Narendra Modi put the number of attacks at 100 million.

Predictably, India's military-industrial infrastructure came in for special attention, while the power ministry confirmed that over 200,000 cyber attacks on the Indian electricity grid were foiled between May 7-10. Their modus operandi comprised the full repertoire of hackers' mischief: website defacements, Denial of Service (DoS) and Distributed Denial of Service (DDoS) attacks (aiming to overwhelm and impair a target server/ network), malware distribution (using viruses to infect systems and gain control) and phishing (use of deceptive emails to extract information). Their cumulative goal: to steal defence information, particularly missile technology, and to undermine vital sectors. Happily, Indian cyber-security agencies, including the Indian Computer Emergency Response Team (CERT-In), the Defence Cyber Agency and the National Critical Information Infrastructure Protection Centre (NCIIPC) successfully thwarted most attacks, with a mere 150 out of the 1.5 million attempts succeeding. Though cyber assets of government institutions, commercial enterprises and the better protected defence organisations attracted 28 per cent, 22 per cent and 17 per cent, respectively, of all attacks, the education (7 per cent), finance (4 per cent) and transport (3 per cent) sectors were not spared. In retaliation, Indian hacking groups took the attack to Pakistan, targeting and breaching critical digital assets of its military and government.

However, India's victory in repulsing these attacks was not absolute. Websites of several Indian military, defence production and defence research insti-

EYES OF THE STORM

Since Pahalgam, foreign hacker groups executed over a million attacks on Indian cyberspace. Here are the sectors that bore the brunt of the barrage



Source: Innefu Labs

1.5 MILLION

Number of hacking attempts on India from after the Pahalgam attack till May 10 by Pakistani and other groups, according to the government. Only 150 succeeded

Illustration by TANMOY CHAKRABORTY